A Research-Oriented Framework for Data-Driven Robust Model Predictive Control

Methodology, Implementation, and Comparative Benchmark

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1 Introduction

Model Predictive Control (MPC) is one of the most powerful frameworks for constrained control of dynamical systems. However, classical MPC relies heavily on the availability of an accurate mathematical model, which may be unavailable, unreliable, or too costly to obtain in many practical applications.

Data-driven MPC (DD-MPC) aims to address this limitation by constructing predictive control laws directly from measured input-output data. Within this paradigm, several different philosophies emerge. Some approaches first identify a predictive model and then apply MPC to the identified model. Some use a learned nonlinear mapping as a surrogate predictive model. Some construct predictions directly from stored trajectories without explicit model identification. Others embed uncertainty directly into the predictive structure and optimize against admissible model realizations.

This document presents a research-oriented framework that includes four representative methods implemented in this repository:

- **DD-RMPC**: Data-Driven Robust Model Predictive Control,
- **KFDDMPC**: Kalman-filter-based dependent-identification DD-MPC,
- **NNDDMPC**: neural-network-based learning-driven DD-MPC,
- **DDMPCA**: data-based predictive control using trajectory-space representation.

The main emphasis of this project is on the DD-RMPC methodology, whose distinguishing feature is the simultaneous treatment of data consistency, model uncertainty, and control synthesis within a structured nonconvex optimization framework.

2 Benchmark Nonlinear System

The prototype compares all methods on the nonlinear discrete-time system

$$\begin{aligned} x_1^+ &= 0.1x_1 + x_1x_2, \\ x_2^+ &= 0.5x_2 - \sin(x_1) + \cos(0.1x_2)u, \\ y &= x_1 + x_2. \end{aligned} \tag{1}$$

The control objective is to track a prescribed reference output while satisfying input, output, and input-rate constraints of the form

$$u_{\min} \le u_k \le u_{\max}, \qquad y_{\min} \le y_k \le y_{\max}, \qquad |u_k - u_{k-1}| \le \Delta u_{\max}. \tag{2}$$

This benchmark is not intended to reproduce all manuscript-level numerical studies. Rather, it serves as a common nonlinear simulation environment for illustrating the qualitative differences between the implemented DD-MPC paradigms.

3 Data-Driven Robust MPC (DD-RMPC)

3.1 Philosophical Foundation

The DD-RMPC framework departs fundamentally from classical identification-based MPC. Instead of estimating a single nominal model from data and then optimizing control actions for that model, DD-RMPC constructs a *set of models* that are consistent with the observed input-output data and explicitly incorporates this uncertainty into the control design.

The central principle is:

> *Replace single-model identification by data-consistent model sets, and compute control actions that remain effective with respect to all admissible models.*

This viewpoint leads to a robust data-driven control philosophy in which prediction and optimization are treated simultaneously rather than sequentially.

3.2 Data-Driven Predictive Representation

Let $\{u_k, y_k\}$ denote the measured input-output sequence. In the proposed DD-RMPC framework, the predictive representation is based on an uncertain autoregressive moving-average structure. More precisely, the future output is described through a UARMA-type model in which the coefficients belong to a data-consistent uncertainty set. At a conceptual level, this can be viewed through the relation

$$y_k = \sum_{i=1}^{n_a} a_i y_{k-i} + \sum_{j=0}^{n_a} b_j u_{k-j}, \tag{3}$$

but unlike classical ARX or ARMA identification, the coefficient vector is not uniquely fixed. Instead, it is embedded in a structured uncertainty description extracted from the recorded data.

In the full DD-RMPC formulation, the uncertainty is represented through:

- a set of admissible coefficient realizations,
- lifted future-output variables associated with uncertainty realizations,
- fitting constraints over a past data window,
- robust tracking constraints over a future prediction horizon.

Hence, the predictive object is not a single trajectory, but a family of trajectories indexed by admissible model realizations.

3.3 Role of Ω_w

The proposed DD-RMPC philosophy is closely related to the set-based view introduced through the family Ω_w, where w denotes the considered window length. Informally, Ω_w contains those systems whose consecutive input-output trajectories over windows of length w can be represented by a suitable uncertain ARMA structure. This viewpoint provides a trade-off between generality and conservativeness:

$$\Omega_1 = \Gamma, \qquad \Omega_\infty = \Gamma_L, \tag{4}$$

where Γ denotes the class of all discrete systems and Γ_L denotes the class of discrete LTI systems.

Thus, w plays the role of a conservativeness parameter. Small values of w increase generality but weaken guaranteed structure, whereas large values of w recover classical LTI-type assumptions. The proposed DD-RMPC philosophy focuses on intermediate values of w, thereby avoiding the extreme conservativeness of purely LTI assumptions while still retaining enough structure to derive meaningful guarantees.

3.4 Robust MPC Formulation

The purpose of DD-RMPC is to compute an input sequence that jointly:

- fits an uncertain predictive model to the recent input-output data,
- propagates admissible future trajectories of that model,
- minimizes the worst-case tracking deviation over the prediction horizon.

To this end, the optimization problem contains:

- decision variables associated with uncertain model coefficients,
- decision variables for future control inputs,
- lifted future-output variables associated with uncertainty vertices,
- a scalar variable γ representing the worst-case fitting/tracking error.

A generic representation is

$$\begin{aligned}
\min \quad & \gamma \\
\text{s.t.} \quad & \text{past data are fitted by an admissible uncertain model,} \\
& \text{future outputs satisfy the continuation equations of that model,} \\
& |y_{k+t}^{(i)} - y_{\text{ref}}| \le \gamma, \\
& u_{\min} \le u_{k+t} \le u_{\max}, \\
& |u_{k+t} - u_{k+t-1}| \le \Delta u_{\max}, \\
& y_{k+t}^{(i)} \in \Omega_y, \qquad u_{k+t} \in \Omega_u.
\end{aligned} \tag{5}$$

This formulation differs from identification-based methods because model fitting and control are not decoupled into two independent steps.

3.5 Quadratic Constraint Structure

The interaction between unknown coefficients, lifted future outputs, and fitting/continuation constraints naturally leads to nonconvex quadratic relations. After collecting all scalar decision variables into a single vector v, the constraints admit a canonical representation of the form

$$v^\top Q_i v + q_i^\top v + r_i \le 0, \qquad i = 1, \dots, m, \tag{6}$$

supplemented by linear equalities and inequalities.

A generic form of the aggregated decision vector is

$$v = \big(\text{model parameters},\, \theta,\, u_f,\, y_f,\, \gamma\big). \tag{7}$$

Therefore, the DD-RMPC problem is cast as a quadratically constrained feasibility/optimization problem with a linear objective and nonconvex quadratic constraints.

3.6 IEOS-Based Solution Strategy

Rather than applying standard convex relaxations such as SDP or SOCP, the problem is treated through a geometric feasibility framework inspired by the Inside–Ellipsoid Outside–Sphere (IEOS) methodology.

The solution procedure consists of three stages.

1. Reformulation. Each quadratic constraint is transformed into an IEOS-compatible structured representation.

2. Iterative Feasibility Update. An iterative map is applied:

$$v^{(\ell+1)} = \mathcal{T}\big(v^{(\ell)}\big), \tag{8}$$

where the operator $\mathcal{T}$ is constructed from the geometry of the transformed feasibility problem.

3. Solution Recovery. Once convergence is reached, the feasible point is mapped back to the original decision variables and the first element of the control sequence is applied.

The important feature of this approach is that it does not rely on generic convex relaxation machinery, but instead exploits the specific structure of the problem.

3.7 Connection to Implementation

The MATLAB implementation directly reflects this formulation. In particular, the code constructs:

- decision variables associated with uncertain coefficients,
- future control variables,
- future lifted outputs,
- a worst-case performance variable,
- quadratic constraint matrices together with their linear terms.

The computational pipeline implemented in the repository follows

$$\texttt{QCFP construction} \longrightarrow \texttt{IEOS mapping} \longrightarrow \texttt{iterative feasibility update} \longrightarrow \texttt{control extraction}. \tag{9}$$

3.8 Interpretation

DD-RMPC can be interpreted as a data-driven robust predictive controller that embeds uncertainty directly into a structured optimization problem and treats model fitting and control synthesis simultaneously. This distinguishes it from identification-based methods, learning-based predictors, and purely trajectory-based schemes.

4 KFDDMPC: Dependent Identification-Based DD-MPC

4.1 Position within DD-MPC Framework

Within the DD-MPC taxonomy, KFDDMPC belongs to the class of *dependent identification approaches*. In this philosophy, a predictive model is first estimated from the measured data and is then used within an otherwise classical MPC framework.

Thus, the control design follows a two-stage procedure:

1. identify a predictive model from recent input-output data,
2. optimize the control sequence for the identified model.

4.2 Online Model Identification

At each time step, the system is approximated by a linear ARX model of the form

$$y_k = \sum_{i=1}^{n_a} a_i y_{k-i} + \sum_{j=1}^{n_b} b_j u_{k-j}. \tag{10}$$

The parameter vector is

$$\theta_k = \begin{bmatrix} a_1 & \cdots & a_{n_a} & b_1 & \cdots & b_{n_b} \end{bmatrix}^\top. \tag{11}$$

In the present implementation, this vector is updated recursively using a Kalman-filter or recursive-least-squares type mechanism. Thus, the model evolves online as new measurements become available.

4.3 Predictive Model

Using the current estimate θ_k, future outputs are predicted through

$$\hat{y}_{k+t} = \sum_{i=1}^{n_a} a_i^{(k)} \hat{y}_{k+t-i} + \sum_{j=1}^{n_b} b_j^{(k)} u_{k+t-j}. \tag{12}$$

The crucial point is that the future trajectory is generated from a single nominal identified model, not from a set of admissible models.

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4.4 MPC Optimization Problem

The control sequence is obtained by solving the quadratic optimization problem

$$\begin{aligned} \min_{\{u_{k+t}\}_{t=0}^{N-1}} \quad & \sum_{t=1}^{N} (\hat{y}_{k+t} - y_{\text{ref}})^2 + \lambda \sum_{t=0}^{N-1} u_{k+t}^2 \\ \text{s.t.} \quad & u_{\min} \le u_{k+t} \le u_{\max}, \qquad t = 0, \dots, N-1, \\ & |u_{k+t} - u_{k+t-1}| \le \Delta u_{\max}, \qquad t = 0, \dots, N-1. \end{aligned} \tag{13}$$

This problem is convex and computationally efficient.

4.5 Structural Characteristics

The dependent identification framework exhibits the following properties:

- model identification and control optimization are carried out sequentially,
- control is based on a single identified nominal model,
- uncertainty in model estimation is not explicitly propagated into the MPC problem.

4.6 Theoretical Implications

The separation between identification and control has several consequences.

- The overall procedure is generally suboptimal, because model fitting and tracking performance are optimized in separate stages.
- The identification step is independent of the control objective.
- Robustness and stability guarantees are difficult to establish in the absence of further structural assumptions on the plant.

4.7 Interpretation

KFDDMPC can be interpreted as a data-driven realization of classical MPC in which the unknown system is replaced online by an identified surrogate model. It is therefore a natural baseline for comparison with DD-RMPC, where model fitting and control are not separated.

5 NNDDMPC: Learning-Based Data-Driven MPC

5.1 Position within DD-MPC Framework

NNDDMPC belongs to the learning-based class of DD-MPC methods. In this philosophy, the predictive model is not obtained through a structured ARX identification and not through direct trajectory-space parameterization. Instead, a nonlinear function approximator is trained from data to represent the input-output map of the system.

Thus, the procedure consists of:

1. learning a nonlinear predictive map from data,
2. embedding that learned map inside an MPC optimization problem.

5.2 Neural-Network Predictive Model

The predictive model is represented by a neural network of the form

$$y_k = \mathcal{N}\left(y_{k-1}, \ldots, y_{k-n_a}, u_{k-1}, \ldots, u_{k-n_b}; \theta\right), \tag{14}$$

where θ contains the weights and biases of the network.

The parameter vector θ is obtained from data by minimizing a prediction loss:

$$\min_{\theta} \sum_{k} \left(y_k - \mathcal{N}(\cdot;\theta)\right)^2. \tag{15}$$

Depending on the implementation, training may be offline, online, or hybrid.

5.3 Multi-Step Prediction Structure

In MPC, the model must generate not only one-step predictions, but an entire future output sequence over the prediction horizon.

The one-step prediction is

$$\hat{y}_{k+1} = \mathcal{N}\left(y_k, \ldots, y_{k-n_a+1}, u_k, \ldots, u_{k-n_b+1}; \theta\right). \tag{16}$$

For later time steps, true future outputs are unavailable. Therefore, the learned model must be recursively reapplied:

$$\hat{y}_{k+t} = \mathcal{N}\left(\hat{y}_{k+t-1}, \ldots, \hat{y}_{k+t-n_a}, u_{k+t-1}, \ldots, u_{k+t-n_b}; \theta\right), \qquad t \geq 2. \tag{17}$$

This recursive structure means that each future prediction depends on previously predicted outputs. As a result, multi-step prediction is obtained through repeated composition of the nonlinear map $\mathcal{N}$.

This has two important consequences. First, the predicted outputs become highly nonlinear functions of the future control inputs. Second, any modeling error introduced at early prediction steps may propagate and amplify along the horizon.

5.4 MPC Optimization Problem

At each time step, the control sequence is obtained by solving

$$\begin{aligned}
\min_{\{u_{k+t}\}_{t=0}^{N-1}} \quad & \sum_{t=1}^{N} \left(\hat{y}_{k+t} - y_{\text{ref}}\right)^2 + \lambda \sum_{t=0}^{N-1} u_{k+t}^2 \\
\text{s.t.} \quad & \hat{y}_{k+t} = \mathcal{N}\left(\hat{y}_{k+t-1}, \ldots, \hat{y}_{k+t-n_a}, u_{k+t-1}, \ldots, u_{k+t-n_b}; \theta\right), \\
& u_{\min} \leq u_{k+t} \leq u_{\max}, \qquad t = 0, \ldots, N-1, \\
& |u_{k+t} - u_{k+t-1}| \leq \Delta u_{\max}, \qquad t = 0, \ldots, N-1.
\end{aligned} \tag{18}$$

Because the predictions are generated recursively by a nonlinear network, this optimization problem is generally nonconvex.

5.5 Structural Characteristics

The learning-based framework has the following main properties:

- strong nonlinear modeling capability,
- implicit model representation through a trained function,
- recursively coupled multi-step predictions.

5.6 Theoretical Implications

The use of a neural predictor leads to several important implications.

- The resulting MPC problem is nonlinear and nonconvex.
- The controller depends strongly on training data, architecture, and fitting quality.
- Stability, recursive feasibility, and robustness are generally difficult to guarantee without additional assumptions or regularization mechanisms.
- Prediction errors may accumulate over long horizons due to recursive rollout.

5.7 Interpretation

NNDDMPC can be interpreted as a flexible nonlinear DD-MPC framework in which dynamics are approximated by a learned function and control actions are computed through nonlinear optimization. It prioritizes modeling flexibility over explicit analytical structure.

6 DDMPCA: Data-Based Predictive Control

6.1 Position within DD-MPC Framework

DDMPCA belongs to the class of non-dependent identification approaches. In this class, predictions are constructed directly from measured data without introducing an explicit intermediate parametric model.

This philosophy is closely related to the behavioral viewpoint, where the system is represented through the set of trajectories consistent with measured input-output data.

6.2 Data-Driven Representation

Let an input-output trajectory of the system be available. Using structured data matrices constructed from this trajectory, admissible future trajectories are represented as linear combinations of recorded trajectories:

$$\begin{aligned} Y_f &= Y_f^{\text{data}}\alpha, \\ U_f &= U_f^{\text{data}}\alpha, \end{aligned} \tag{19}$$

where α is a coefficient vector that parameterizes feasible trajectories.

6.3 Consistency with Measured Data

To ensure compatibility with the current operating point, the recent input-output measurements must be matched:

$$\begin{aligned} Y_p^{\text{data}}\alpha &= y_{\text{past}}, \\ U_p^{\text{data}}\alpha &= u_{\text{past}}. \end{aligned} \tag{20}$$

These equations enforce that the predicted trajectory belongs to the trajectory space compatible with both the data and the current measurements.

6.4 MPC Optimization Problem

The control problem is formulated as

$$
\begin{aligned}
\min_{\alpha} \quad & \sum_{t=1}^{N} \left((Y_f)_t - y_{\text{ref}}\right)^2 + \lambda \|\alpha\|^2 \\
\text{s.t.} \quad & Y_f = Y_f^{\text{data}} \alpha, \\
& U_f = U_f^{\text{data}} \alpha, \\
& Y_p^{\text{data}} \alpha = y_{\text{past}}, \\
& U_p^{\text{data}} \alpha = u_{\text{past}}, \\
& u_{\min} \leq (U_f)_t \leq u_{\max}, \\
& |(U_f)_t - (U_f)_{t-1}| \leq \Delta u_{\max}.
\end{aligned}
\tag{21}
$$

The optimal control action is the first element of the optimal future input sequence.

6.5 Structural Characteristics

The key structural features of DDMPCA are:

- no explicit model identification,
- linear parameterization of trajectories through α,
- typically convex quadratic optimization.

6.6 Theoretical Implications

The effectiveness of DDMPCA depends heavily on the quality and richness of the stored data.

- The available data must sufficiently excite the system dynamics.
- The method is closely tied to trajectory-space representations such as those arising from behavioral system theory.
- Performance may degrade under noise, insufficient excitation, or mismatch with the underlying assumptions.

6.7 Interpretation

DDMPCA may be interpreted as selecting, among all trajectories consistent with measured data, the one that best satisfies the control objective. It avoids explicit identification, but places the main burden on data richness and structural assumptions.

7 Comparative Perspective

The four methods represent fundamentally different DD-MPC paradigms. Their main differences stem from how they represent system behavior and how they treat uncertainty.

7.1 Philosophical Classification

A useful classification is as follows:

- **Dependent identification approaches:** a model is first estimated from data and then used for control, as in KFDDMPC;
- **Non-dependent identification approaches:** predictions are constructed directly from data without explicit model identification, as in DDMPCA;

- **Learning-based approaches:** a nonlinear mapping is learned from data and embedded in MPC, as in NNDDMPC;
- **Robust set-based approaches:** a family of data-consistent models is constructed and control is optimized against admissible realizations, as in DD-RMPC.

7.2 Fundamental Trade-Off

A central observation in DD-MPC is the trade-off between generality and guarantees. If a method makes almost no assumption on the system, then it may be widely applicable but theoretical guarantees become difficult. If stronger structure is assumed, then guarantees become more tractable but applicability becomes restricted. The four methods reflect different positions in this trade-off.

7.3 Method Comparison

Method	Modeling Philosophy	Optimization	Main Limitation
DD-RMPC	Robust set-based data-driven	Nonconvex QCQP / IEOS	Higher computational complexity
KFDDMPC	Dependent identification	Convex QP	Separation-induced suboptimality
NNDDMPC	Learned nonlinear predictor	Nonconvex NLP	Limited guarantees and error propagation
DDMPCA	trajectory-based	Convex QP	Strong dependence on excitation

7.4 Interpretation of Differences

KFDDMPC separates identification and control, which preserves convexity but may introduce suboptimality. NNDDMPC increases modeling flexibility through nonlinear learning, but the resulting optimization is nonconvex and theoretically harder to analyze. DDMPCA eliminates explicit model identification and operates directly in trajectory space, but its success depends strongly on data richness and structural assumptions such as those underlying behavioral representations. DD-RMPC differs from all of them by explicitly embedding uncertainty into the optimization problem and optimizing against admissible model realizations.

7.5 Key Insight

The essential difference between the four methods lies in how uncertainty is treated:

- KFDDMPC effectively ignores uncertainty after identification,
- NNDDMPC absorbs uncertainty into a learned nonlinear surrogate,
- DDMPCA assumes the available data adequately span the relevant trajectory space,
- DD-RMPC represents uncertainty explicitly through a structured admissible set and incorporates it directly into control synthesis.

This makes DD-RMPC fundamentally distinct: uncertainty is neither hidden nor ignored, but directly modeled and optimized against.

7.6 Practical Perspective

The present repository provides a prototype-level implementation of the four paradigms together with an illustrative benchmark. The purpose is methodological exposition and comparative understanding rather than full-scale archival benchmarking. Nevertheless, the implementation is structured so that readers can clearly identify the conceptual role played by each method and the distinctive contribution of DD-RMPC.

8 Conclusion

This project presents a structured comparison of four important DD-MPC paradigms and highlights the implementation of a robust data-driven predictive control framework based on uncertain model sets and IEOS-type feasibility ideas.

The resulting document and code serve as a bridge between conceptual methodology, algorithmic structure, and practical implementation. In particular, they clarify why DD-RMPC should be viewed not simply as another predictor-based controller, but as a distinct robust set-based DD-MPC philosophy.